Supplementary Table S1. Forecasting, control, and evaluation settings

The parameter values apply to the 230 MW PV / 195.5 MW modeled POI boundary and the specified 3 h closed-loop study.

Category	Parameter	Setting
Study frame	**Evaluation support**	Twelve chronological monthly test folds from September 2024 through August 2025. All training and validation target valid times precede the start of the corresponding test fold.
Study frame	**Temporal resolution and horizons**	3 h dispatch interval. H = 8 corresponds to a 24 h MPC horizon (current step plus future leads 3-21 h); H = 24 corresponds to a 72 h MPC horizon (current step plus future leads 3-69 h). The 72 h forecast row is retained only for diagnostic provenance.
Study frame	**Matched forecast branches**	History-CatBoost and Weather-CatBoost use the same residual-on- diurnal target, candidate grid, chronological split, cap-focused weights, early stopping, residual shrinkage, matched direct-NWP training support, and fallback rules. Weather-CatBoost differs only by the addition of operational NWP variables.
Study frame	**Matched training support**	Both branches train only on rows satisfying matched_catboost_training_support_flag = 1. This support is the direct operational-NWP support used by the weather branch.
Study frame	**Weather fallback**	When operational NWP is unavailable, Weather-CatBoost uses the exact frozen diurnal forecast. NWP values are not imputed and future weather observations are not substituted.
Input features	**History/calendar/lead features (15)**	run_age_h; lead_from_decision_h; lead_from_run_h; measured_pv_issue_mw; pv_lag_3h_mw; pv_lag_6h_mw; pv_lag_24h_mw; pv_lag_48h_mw; pv_roll_24h_mw; hour_sin; hour_cos; doy_sin; doy_cos; dow_sin; dow_cos.
Input features	**Operational NWP increment (11)**	nwp_ssrd_Wm2; nwp_lwd_Wm2; nwp_t2m_C; nwp_d2m_C; nwp_sp_Pa; nwp_u10_mps; nwp_v10_mps; nwp_ws10_mps; nwp_wd10_deg; nwp_tp_rate_mm_s; nwp_tp_3h_mm.
Input features	**Forecast reconstruction**	For each lead, predicted PV is clip(diurnal reference + lambda × predicted residual, 0, 230 MW). Models are trained separately by lead.
CatBoost model grid	**C01_compact_d4**	Depth 4; learning rate 0.05; maximum 1000 iterations; L2 leaf regularization 5; Bernoulli bootstrap; subsample 0.8; random strength 1.0; rsm 1.0.
CatBoost model grid	**C02_balanced_d6**	Depth 6; learning rate 0.05; maximum 1000 iterations; L2 leaf regularization 5; Bernoulli bootstrap; subsample 0.8; random strength 1.0; rsm 1.0.
CatBoost model grid	**C03_regularized_d6_slow**	Depth 6; learning rate 0.03; maximum 1500 iterations; L2 leaf regularization 8; Bernoulli bootstrap; subsample 0.8; random strength 1.0; rsm 1.0.
CatBoost training	**Common fixed controls**	CatBoostRegressor; loss_function = RMSE; eval_metric = RMSE; task_type = CPU; thread_count = 1; random_seed = 42; nan_mode = Min.
CatBoost training	**Chronological validation**	Preferred validation set: final 60 days of the pretest period. Prespecified fallback: chronological 75% training / 25% validation when the fixed tail violates the minimum sample counts. Minimums: 40 training rows and 12 validation rows.
CatBoost training	**Early stopping**	50 rounds, evaluated on the chronological validation tail using weighted RMSE on the residual target; validation weights are applied.

Category	Parameter	Setting
CatBoost training	**Cap-focused sample weights**	Ordinary row = 1; daylight row = 2; true over-cap row = 10. The policy was frozen before test outcomes.
CatBoost training	**Candidate and shrinkage selection**	Lexicographic order: lower validation over-cap false-negative rate; lower daylight MAE; lower cap-weighted RMSE; fewer selected boosting iterations; deterministic candidate identifier. Residual shrinkage lambda is selected from {0, 0.25, 0.5, 0.75, 1, 1.25}.
CatBoost training	**Refit and test isolation**	After selection, the model is refitted on all matched pretest rows with the selected iteration count and lambda locked. The selection/refit lock is hashed before the read-only test evaluation; test targets, errors, and controller outcomes are excluded from selection.
CatBoost training	**Campaign scale**	12 folds × 23 controller-consumed leads × 2 branches × 3 candidates = 1,656 candidate fits; 552 selected refits; 133,676 locked CatBoost test predictions; 266,784 rows in the common frozen forecast table.
CatBoost training	**Forecast freeze**	Diurnal, History-CatBoost, Weather-CatBoost, and Oracle trajectories are written, hashed, and reloaded before forecast metrics or MPC are computed. Compressed forecast SHA-256: e5968c2996691ae5ee189e30190f446731f2561a0f652cfee7df065da69 4e8a9.
HGB robustness grid	**H01_compact_d4**	max_depth = 4; learning_rate = 0.05; max_iter = 200; l2_regularization = 5; max_leaf_nodes = 31; min_samples_leaf = 20; max_bins = 255.
HGB robustness grid	**H02_balanced_d6**	max_depth = 6; learning_rate = 0.05; max_iter = 300; l2_regularization = 5; max_leaf_nodes = 31; min_samples_leaf = 20; max_bins = 255.
HGB robustness grid	**H03_regularized_d6_slow**	max_depth = 6; learning_rate = 0.03; max_iter = 500; l2_regularization = 8; max_leaf_nodes = 31; min_samples_leaf = 20; max_bins = 255.
HGB robustness	**Common fixed controls**	HistGradientBoostingRegressor with squared-error loss; random_state = 42; validation_fraction = None; n_iter_no_change = 30; tolerance = 1e-7. The same feature ablation, sample weights, validation rules, shrinkage grid, and exact diurnal fallback are used as in the CatBoost comparison.
HGB robustness	**External early stopping and refit**	The chronological validation tail is supplied explicitly; no internal random validation split is used. The selected n_iter_ is locked and then used for a pretest refit with early stopping disabled.
MPC and physical model	**Plant and export boundary**	PV rated power = 230 MW; export-cap fraction = 0.85; export cap = 195.5 MW; active-power model; no grid import for battery charging.
MPC and physical model	**Nominal BESS**	Charge/discharge power limits = 30/30 MW; energy capacity = 140 MWh; charge/discharge efficiencies = 0.95/0.95; physical SOC range = 10%-90% (14-126 MWh).
MPC and physical model	**Objective weights**	Curtailment weight = 1000; battery-throughput regularization weight = 1; terminal-energy deviation weight = 50.
MPC and physical model	**Operational constraints**	MILP power balance; grid export constrained to 0-195.5 MW; binary charge/discharge exclusivity; charging bounded by available PV; discharging bounded by forecasted export headroom; first optimized action only is executed before state update.
MPC and physical model	**Solver and tolerances**	scipy.optimize.milp with the HiGHS backend; relative MIP-gap option = 1e-8; numerical tolerance = 1e-6; terminal tolerance = 1e-5 MWh; solver success required. Nominal campaign: no explicit time limit. BESS-sensitivity campaign: 5 s time limit per solve.
Terminal fairness	**Initial and terminal condition**	Every monthly fold begins and ends at exactly 50% SOC. For the nominal 140 MWh BESS, E0 = ET = 70 MWh.

Category	Parameter	Setting
Terminal fairness	**Steering window**	168 h prebuffer (56 three-hour intervals) followed by a 24 h endgame (8 intervals); total window = 192 h (64 intervals).
Terminal fairness	**Causal steering schedule**	Seven equal 8 MWh lower-floor increments during the prebuffer, scheduled at 19:00 Asia/Baku, reaching 70 MWh at the start of the endgame.
Terminal fairness	**Common terminal forecast**	During the shared prebuffer and endgame, all compared methods use the same causal diurnal forecast for terminal steering; current PV remains measured. This prevents forecast-dependent terminal reachability from biasing the comparison.
Terminal fairness	**Fold-end enforcement**	When the true fold endpoint lies inside the nominal horizon, the horizon is shortened to the fold boundary and ET = target is imposed exactly with backward reachability.
BESS sensitivity	**Power axis**	P_BESS = {15, 30, 45, 60} MW with E_BESS fixed at 140 MWh.
BESS sensitivity	**Energy axis**	E_BESS = {70, 140, 210, 280} MWh with P_BESS fixed at 30 MW. The nominal 30 MW/140 MWh point is executed once and shared by both axes.
BESS sensitivity	**Forecast handling**	The authenticated CatBoost forecast table is reused throughout the BESS-sensitivity analysis without model retraining. Initial and final SOC remain at 50% for every capacity.
Statistical analysis	**Monthly pairing**	Twelve ordered monthly folds; Weather-CatBoost is paired against History-CatBoost and Diurnal. Positive benefit means Weather- CatBoost is better for the directioned metric.
Statistical analysis	**Bootstrap**	Circular moving-block bootstrap over ordered folds; block length = 3 months; 10,000 replicates; random seed = 20260802.
Statistical analysis	**Secondary tests and multiplicity**	Two-sided Wilcoxon signed-rank test and exact sign test. BESS- sensitivity p-values use Benjamini-Hochberg adjustment. Tie tolerance = 1e-6 MWh.
Statistical analysis	**Near-cap definition**	Absolute difference between measured PV and the 195.5 MW cap not exceeding 10% of the cap: |PV - cap| <= 19.55 MW.
Software and execution environment	**CatBoost forecast campaign**	Python 3.13.5; CatBoost 1.2.8; NumPy 2.3.5; pandas 2.2.3; Linux 6.12.13 x86_64 with glibc 2.41.
Software and execution environment	**HGB robustness and shared-terminal control**	Python 3.13.5; scikit-learn 1.8.0; joblib 1.5.3; NumPy 2.3.5; pandas 2.2.3; SciPy 1.17.0; openpyxl >= 3.1.
Software and execution environment	**Recorded nominal MILP solutions and hardware**	23,336 optimal MILP solves: mean 1.449 ms; median 1.415 ms; 95th percentile 2.227 ms; maximum 41.118 ms. Maximum relative MIP gap 9.96e-11; maximum branch-and-bound nodes 1; solver-call timings only. Linux (kernel 6.12.13, glibc 2.41); AMD EPYC 9V74; four allocated CPU cores; 4 GiB memory limit. MILP problem sizes: 24 h 43 variables (8 binary); 72 h 123 variables (24 binary).